

Technology Stack

Date	15 March 2026
Team ID	xxxxxx
Project Name	xxxxxx
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	HTML5, CSS3, JavaScript, Jinja2 Templates	Provides a responsive, user-friendly interface for students to interact with the Edu Genie application.

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python, Fast API, Uvicorn	Handles user requests, processes AI queries, and serves the web application with high performance.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	No Database (Prototype Version)	The application processes user requests in real time and does not store user information.
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Local Host (Uvicorn Server) <i>(or Render if you deployed it)</i>	Used to run and test the application during development. Can be deployed to cloud platforms in the future.
5	Version Control & CI/CD <i>(e.g., GitHub, GitLab, Jenkins)</i>	Git, GitHub	Used for version control, source code management, collaboration, and maintaining project history.

6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	Groq API (Llama 3.1-8B Instant), OpenAI Python SDK, Python-dotenv, Visual Studio Code	Groq API generates AI responses, OpenAI SDK connects to the API, Python-dotenv securely manages API keys, and VS Code is used for development.
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